
Software Requirements Specification

for

SEAL Hackathon Management System, Release 1.0

Version 1.0 approved

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1. Business Process

The following tables organize each SEAL process into process boundaries, lane responsibilities, ordered activities, and decision/exception paths. Each activity row identifies the responsible lane so it can be transferred directly into a swimlane diagram. A dedicated landscape figure page follows every process specification.

1.1 Account Registration and Approval

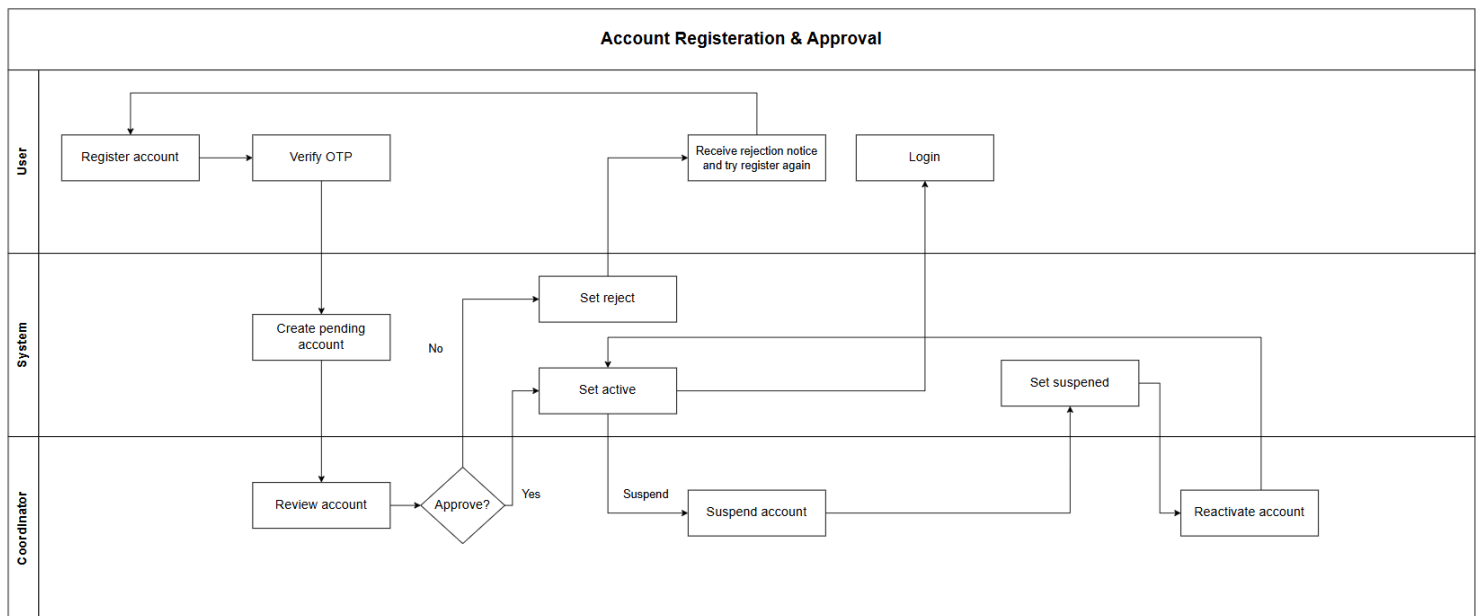


Figure 1.1 Account registration and approval swimlane

Process boundary

Field	Description
Business-flow code	BF-01
Purpose	Move a newly registered account through OTP verification and coordinator review, including rejection/resubmission and suspension/reactivation.
Inputs	Registration information, OTP, coordinator decision, and rejection/suspension reason where applicable.
Start event	The user registers an account and verifies OTP.
End event	The account is Active, or remains Rejected/Suspended until the diagrammed recovery transition occurs.

Outputs	Account state, stored review result/reason, user notification, and login availability for Active accounts.
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Lane responsibilities

Lane / actor	Responsibility
User	Registers, verifies OTP, receives a rejection notice, resubmits, and logs in after activation.
System	Creates Pending Approval and applies Active, Rejected, or Suspended state changes.
Coordinator	Reviews, approves/rejects, suspends, and reactivates accounts.

Main flow

Step	Responsible lane	Activity / expected result
1	User	Registers an account and verifies OTP.
2	System	Creates the account in Pending Approval state.
3	Coordinator	Reviews the pending account and decides Approve?.
4	System	If approved, sets Active and notifies the user; the user can log in.
5	System	If rejected, sets Rejected and stores the reason.
6	User	Receives the rejection notice, corrects the registration, and resubmits.
7	System	Clears the rejection reason and returns the account to Pending Approval.
8	Coordinator	May suspend an Active account; the system sets Suspended.
9	Coordinator	May reactivate a Suspended account; the system returns it to Active.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Approve?	Yes: Pending Approval -> Active. No: Pending Approval -> Rejected.
Rejected account	Resubmission clears the reason and returns the account to Pending Approval.
Suspended account	Normal Active access resumes only after coordinator reactivation.
OTP prerequisite	The system creates Pending Approval only after the OTP-verification activity.

1.2 Event Configuration

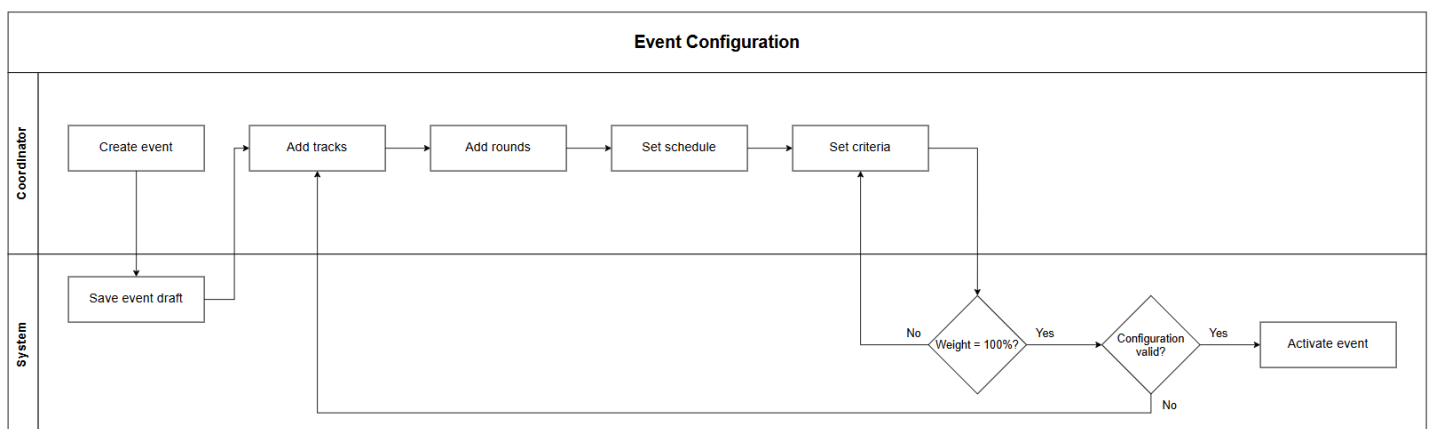


Figure 1.2 Event lifecycle swimlane

Process boundary

Field	Description
Business-flow code	BF-02
Purpose	Create an event Draft, configure tracks, rounds, schedule and criteria, and activate it only after both validation gateways pass.

Inputs	Event definition, tracks, rounds, schedule, scoring criteria, and criterion weights.
Start event	The coordinator creates an event and the system saves it as Draft.
End event	A valid Draft is activated as Ongoing; later closure/cancellation moves it to Ended.
Outputs	Persisted event configuration and an Event state of Draft, Ongoing, or Ended.

Lane responsibilities

Lane / actor	Responsibility
Coordinator	Creates the event and configures tracks, rounds, schedule, and scoring criteria.
SEAL System	Saves the Draft, validates criterion weight and overall configuration, and activates the event.

Main flow

Step	Responsible lane	Activity / expected result
1	Coordinator	Creates the event.
2	System	Saves the event as Draft.
3	Coordinator	Adds tracks.
4	Coordinator	Adds rounds.
5	Coordinator	Sets the schedule.
6	Coordinator	Sets scoring criteria.
7	System	Checks Weight = 100%?.
8	System	When the weight is correct, checks Configuration valid?.
9	System	Activates the event and changes Draft to Ongoing when configuration is valid.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Weight = 100%?	No: return to Set criteria. Yes: continue to complete configuration validation.
Configuration valid?	No: return to configuration. Yes: activate the event.
Close or cancel	An Ongoing event becomes Ended through this lifecycle transition.

1.3 Team Formation

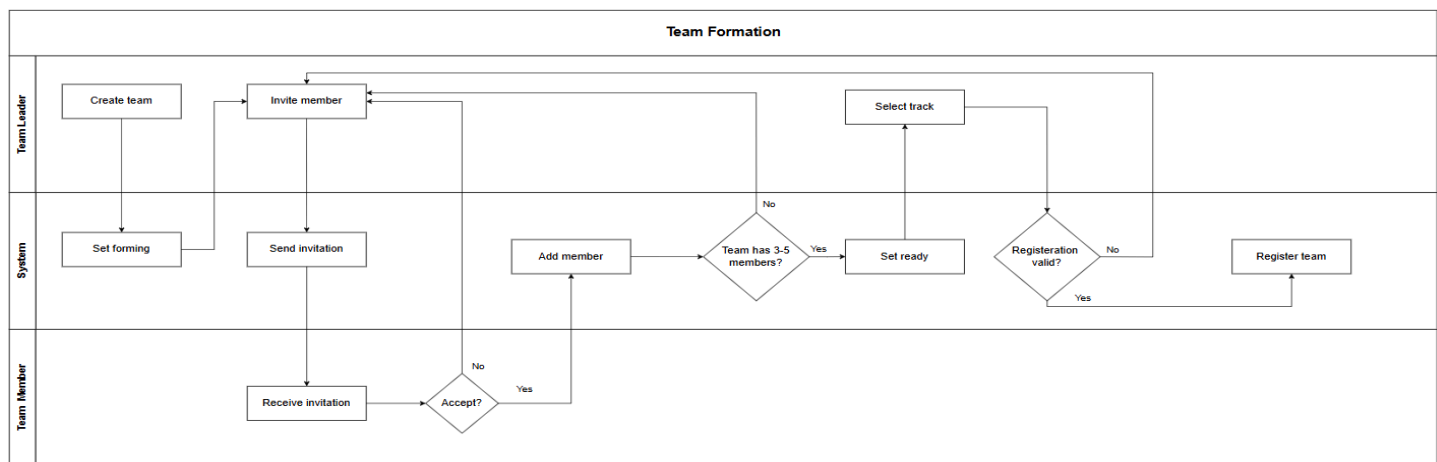


Figure 1.3 Team formation swimlane

Process boundary

Field	Description
Business-flow code	BF-03
Purpose	Create an invitation-based team, reach the three-to-five-member Ready condition, select a track, and register a valid team.
Inputs	Team creator, member invitations and responses, current member count, selected track, and registration data.
Start event	The Team Leader creates a team; the system sets Forming.
End event	The system registers a Ready team after registration validation, or retains Forming/Disqualified until conditions change.
Outputs	Invitations, team membership, Forming/Ready/Disqualified state, selected track, and valid team registration.

Lane responsibilities

Lane / actor	Responsibility
Team Leader	Creates the team, invites members, selects a track, and initiates registration.
Team Member	Receives and accepts/rejects invitations.
System	Sets Forming/Ready, sends invitations, adds members, validates registration, and registers the team.
Coordinator	May disqualify the team while the event is ongoing.

Main flow

Step	Responsible lane	Activity / expected result
1	Team Leader	Creates a team.
2	System	Sets Forming.
3	Team Leader	Invites a member; the system sends the invitation.
4	Team Member	Receives the invitation and decides Accept?.
5	System	On acceptance, adds the member and checks Team has 3-5 members?.
6	System	If the size condition fails, returns to Invite member; if it passes, sets Ready.
7	Team Leader	Selects a track.
8	System	Checks Registration valid?.
9	System	Registers the team when valid; otherwise returns to the formation path.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Accept?	No: return to Invite member. Yes: add the member.
Team has 3-5 members?	No: remain/return to Forming. Yes: set Ready.
Registration valid?	No: return to team formation. Yes: register the team.

Member removed	Ready -> Forming when membership falls below three.
Disqualification/restoration	Forming or Ready -> Disqualified during an ongoing event; restoration returns to Forming or Ready according to member count.

1.4 Mentor and Judge Assignment

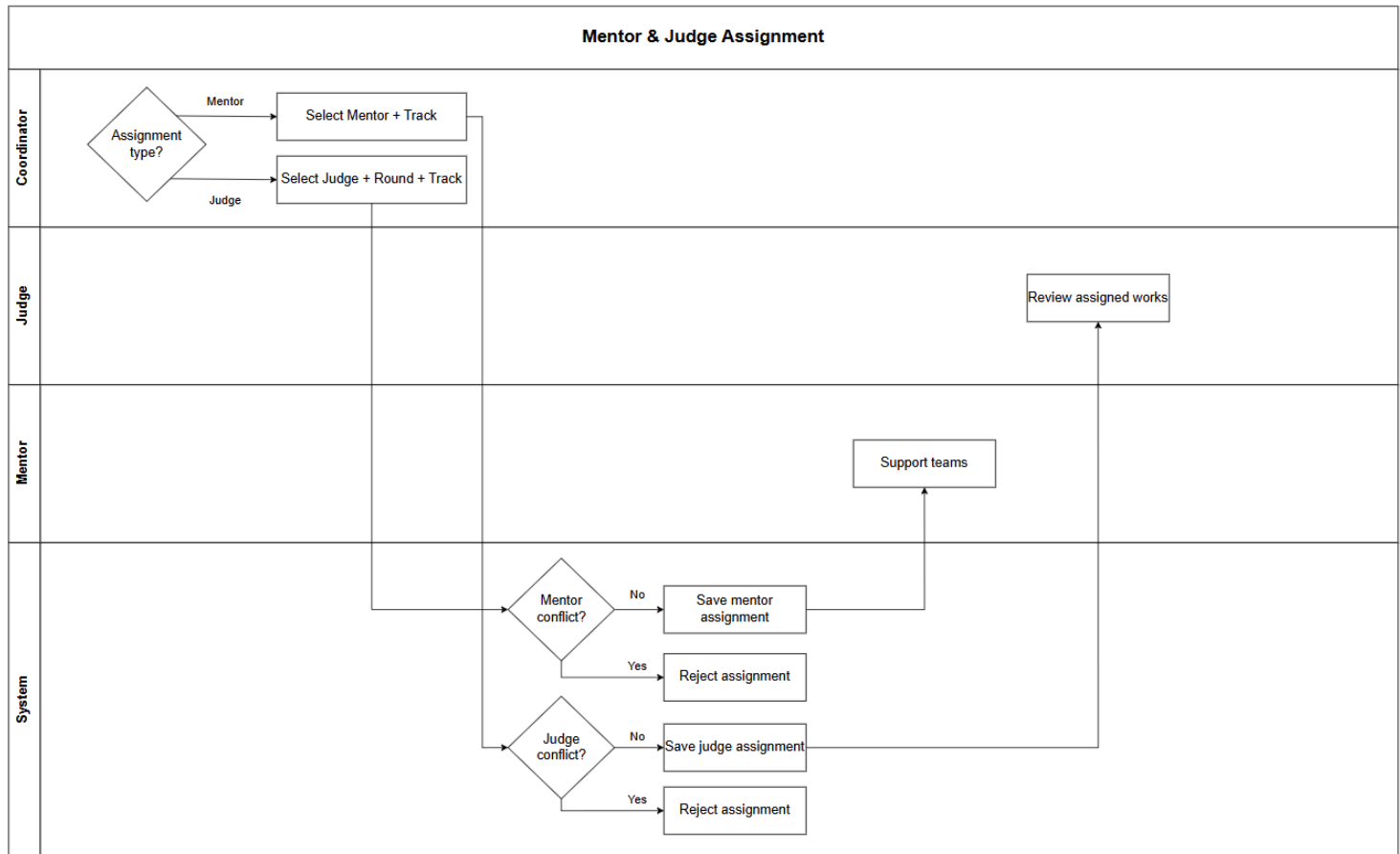


Figure 1.4 Mentor and Judge Assignment Swimlane

Process boundary

Field	Description
Business-flow code	BF-04
Purpose	Assign a Mentor to a track or a Judge to a round/track while preventing assignment conflicts.

Inputs	Assignment type, Mentor or Judge identity, track, round for Judge assignment, active-event status, and conflict result.
Start event	The coordinator selects Assignment type?.
End event	The assignment is Assigned and the staff member can perform the assigned work, or the system rejects it and leaves the record Unassigned.
Outputs	Assigned/Unassigned state, Mentor-track assignment or Judge-round-track assignment, conflict rejection, and staff work access.

Lane responsibilities

Lane / actor	Responsibility
Coordinator	Chooses assignment type and selects the Mentor/Judge with the required track and round.
System	Checks conflicts, saves a valid assignment, or rejects a conflicting assignment.
Mentor	Supports teams after a Mentor + Track assignment is saved.
Judge	Reviews assigned works after a Judge + Round + Track assignment is saved.

Main flow

Step	Responsible lane	Activity / expected result
1	Coordinator	Selects Assignment type?.
2	Coordinator	For Mentor, selects Mentor + Track.
3	System	Checks Mentor conflict?.
4	System	If no conflict, saves the Mentor assignment; the Mentor supports teams.
5	Coordinator	For Judge, selects Judge + Round + Track.
6	System	Checks Judge conflict?.
7	System	If no conflict, saves the Judge assignment; the Judge reviews assigned works.
8	System	Rejects either assignment when the corresponding conflict check is Yes.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Assignment type?	Mentor: require Mentor + Track. Judge: require Judge + Round + Track.
Mentor conflict?	Yes: reject assignment. No: save assignment and enable Mentor support.
Judge conflict?	Yes: reject assignment. No: save assignment and enable Judge review.
Create assignment guard	The state chart requires an active event and no conflict.
Remove assignment guard	Assigned -> Unassigned only while the assignment is unlocked.

1.5 Submission Management

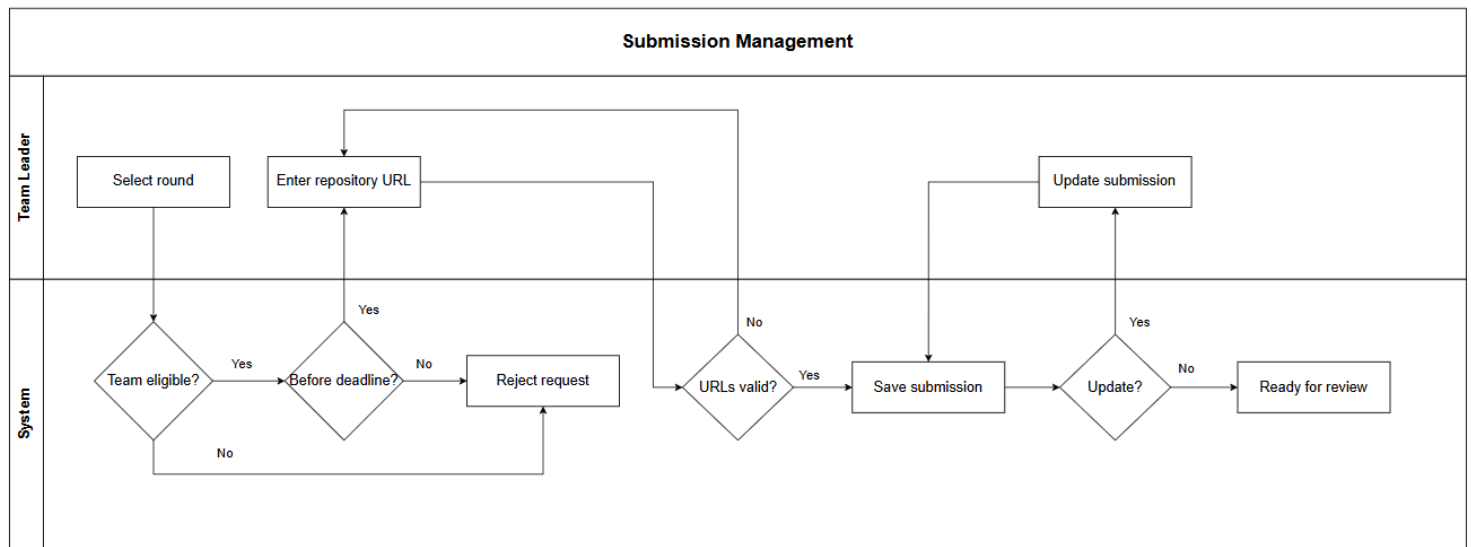


Figure 1.5 Submission Management Swimlane

Process boundary

Field	Description
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Business-flow code	BF-05
Purpose	Validate team eligibility, deadline and URLs before saving a round submission, while allowing pre-deadline updates.
Inputs	Selected round, valid team registration, active submission deadline, repository URL, and update request.
Start event	The Team Leader selects a round.
End event	A valid submission is saved as Submitted and becomes ready for review; Judge scoring hands it off to Evaluating.
Outputs	Rejected request or a timestamped Submitted record, update history through the Submitted self-transition, and Ready for review handoff.

Lane responsibilities

Lane / actor	Responsibility
Team Leader	Selects the round, enters the repository URL, and updates the submission.
System	Checks eligibility/deadline/URLs, rejects invalid requests, saves the submission, and marks it ready for review.

Main flow

Step	Responsible lane	Activity / expected result
1	Team Leader	Selects the round.
2	System	Checks Team eligible?.
3	System	If eligible, checks Before deadline?.
4	Team Leader	Enters the repository URL.
5	System	Checks URLs valid?.
6	System	If valid, saves the submission and timestamp as Submitted.
7	System	Checks Update?.
8	Team Leader	If an update is requested before the deadline, updates the submission; the system validates and saves again.
9	System	If there is no update, marks the submission ready for review.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Team eligible?	No: reject request. Yes: continue to deadline validation.
Before deadline?	No: reject request. Yes: allow URL entry/update.
URLs valid?	No: return to Enter repository URL. Yes: save the submission.
Update?	Yes: update and return to Save submission. No: mark Ready for review.

1.6 Evaluation and Scoring

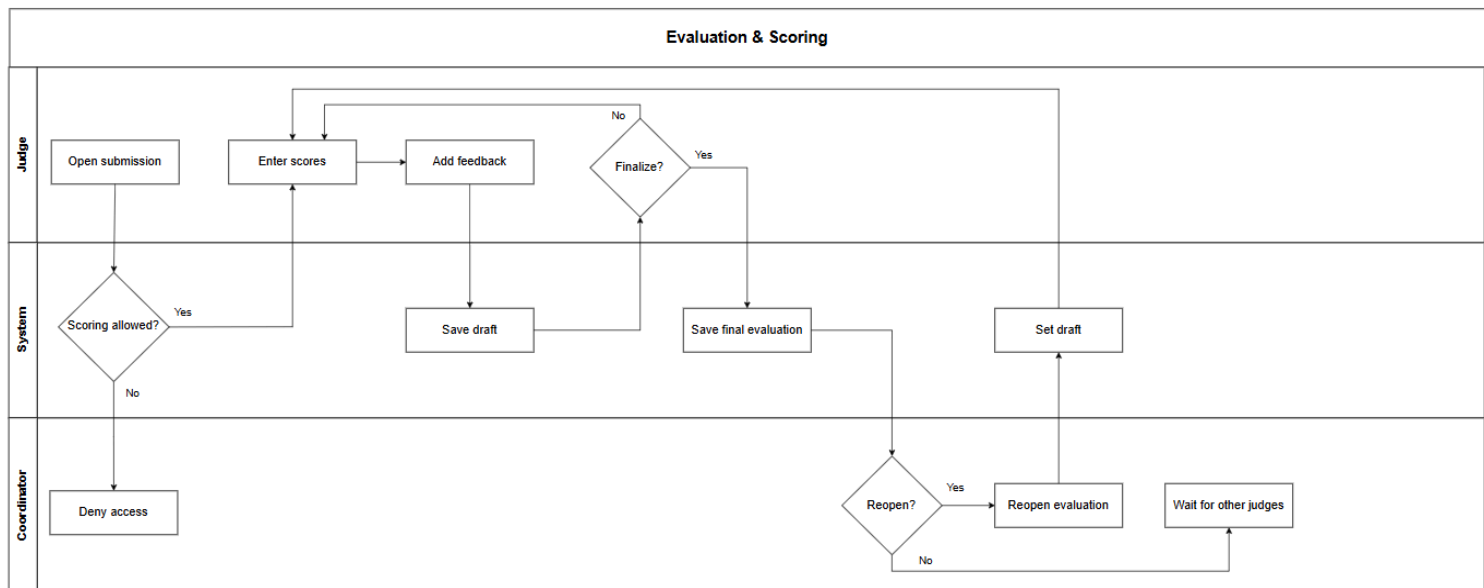


Figure 1.6 Evaluation and Scoring

Process boundary

Field	Description
Business-flow code	BF-06

Purpose	Allow an assigned Judge to score and comment on a submission, save Draft work, finalize a complete evaluation, and support controlled reopening.
Inputs	Assigned submission, scoring-open status, rubric criteria, criterion scores, feedback, and reopen decision.
Start event	The Judge opens an assigned submission; the system checks Scoring allowed?.
End event	The evaluation is Finalized, or remains Draft for more scoring/reopened work; other judges may still be pending.
Outputs	Draft or Finalized Judge evaluation, saved scores, feedback, Evaluating submission handoff, access denial where applicable.

Lane responsibilities

Lane / actor	Responsibility
Judge	Opens the assigned submission, enters scores, adds feedback, and chooses whether to finalize.
System	Checks scoring access, saves Draft/Finalized evaluation state, denies unauthorized scoring, and restores Draft on reopen.
Coordinator	Decides whether a Finalized evaluation should be reopened while the event is not ended.

Main flow

Step	Responsible lane	Activity / expected result
1	Judge	Opens the assigned submission.
2	System	Checks Scoring allowed?.
3	Judge	When allowed, enters scores and adds feedback.
4	System	Saves the evaluation as Draft and changes the submission to Evaluating when scores are saved.
5	Judge	Chooses Finalize?.
6	System	Finalizes only when all criteria are scored and scoring remains open.
7	Coordinator	Chooses Reopen?.

8	System	If reopened while the event is not ended, returns Finalized to Draft; otherwise waits for other judges.
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Decision points and exception flow

Decision / exception	Branch, handling and outcome
Scoring allowed?	No: deny access. Yes: permit score and feedback entry.
Finalize?	No: return to Enter scores. Yes: save Finalized when the state-chart guard is satisfied.
All criteria scored?	If incomplete, the evaluation cannot transition from Draft to Finalized.
Reopen?	Yes only when the event is not ended; Finalized -> Draft. No: wait for other judges.

1.7 Ranking and Promotion

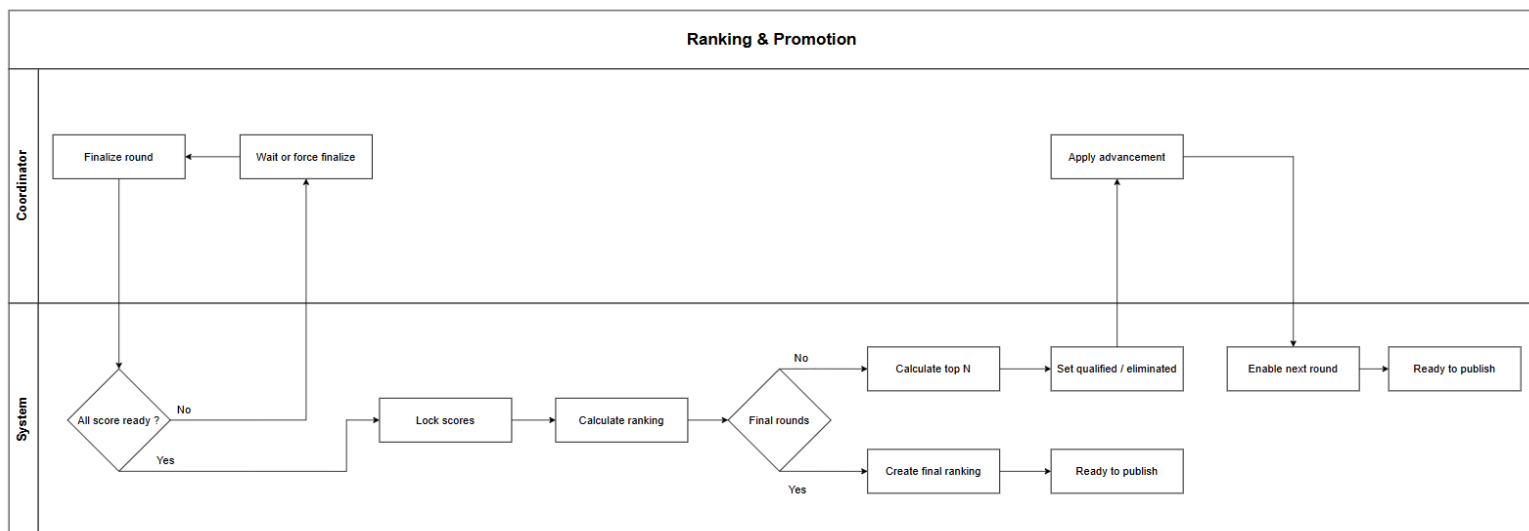


Figure 1.7 Ranking and Promotion

Process boundary

Field	Description
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Business-flow code	BF-07
Purpose	Lock round scoring, create rankings, calculate top-N qualification for non-final rounds, apply advancement, and prepare results for publication.
Inputs	Judge evaluations, score-readiness status, force-finalize decision, final/non-final round type, ranking data, and top N.
Start event	The coordinator selects Finalize round while round processing is ScoringOpen.
End event	The round is ReadyToPublish/Published; non-final teams are Qualified or Eliminated and advancement is applied.
Outputs	ScoringLocked round, ranking, QualificationCalculated state, Qualified/Eliminated teams, AdvancementApplied, enabled next round, and ReadyToPublish result.

Lane responsibilities

Lane / actor	Responsibility
Coordinator	Finalizes the round, selects wait/force-finalize handling, and applies advancement.
System	Checks score readiness, locks scores, calculates ranking/qualification, enables the next round, and prepares results.

Main flow

Step	Responsible lane	Activity / expected result
1	Coordinator	Finalizes the round.
2	System	Checks All score ready?.
3	Coordinator	If not ready, waits or force-finalizes and retries Finalize round.
4	System	When ready, locks scores and calculates ranking.
5	System	Checks Final round?.
6	System	For a non-final round, calculates top N and sets Qualified/Eliminated.
7	Coordinator	Applies advancement.

8	System	Enables the next round and sets ReadyToPublish.
9	System	For the final round, creates the final ranking and sets ReadyToPublish.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
All score ready?	No: wait or force-finalize, then retry. Yes: lock scores.
Finalize-round guard	ScoringOpen -> ScoringLocked when all evaluations are finalized or force-finalize is used.
Final round?	No: calculate top N and apply advancement. Yes: create final ranking.
Within top N?	Yes: Pending -> Qualified. No: Pending -> Eliminated.

1.8 Disqualification and Audit

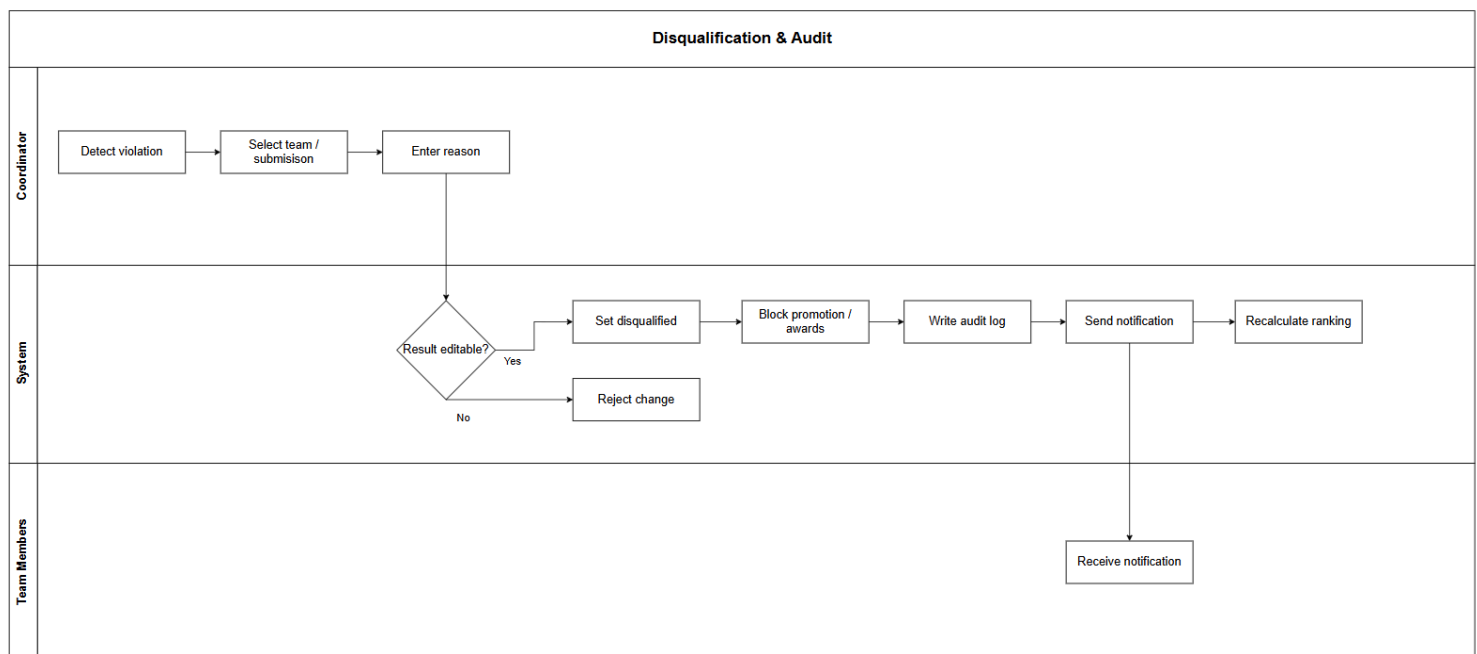


Figure 1.8 Disqualification and Audit

Process boundary

Field	Description
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Business-flow code	BF-08
Purpose	Apply or revoke a coordinator disqualification only while results are editable, block promotion/awards, audit the action, notify teams, and recalculate ranking.
Inputs	Detected violation, selected team/submission, reason, current ranking state, and Result editable? status.
Start event	The coordinator detects a violation.
End event	The result is rejected as non-editable, or the entry is Disqualified with audit/notification/recalculation completed; revocation returns it to Pending.
Outputs	Disqualified/Pending qualification state, blocked promotion/awards, audit record, team notification, and recalculated ranking.

Lane responsibilities

Lane / actor	Responsibility
Coordinator	Detects a violation, selects the team/submission, enters a reason, and may revoke disqualification.
System	Checks editability, sets Disqualified, blocks promotion/awards, writes audit log, notifies, and recalculates.
Team Members	Receive the disqualification notification.

Main flow

Step	Responsible lane	Activity / expected result
1	Coordinator	Detects a violation.
2	Coordinator	Selects the team/submission.
3	Coordinator	Enters the reason.
4	System	Checks Result editable?.
5	System	If editable, sets Disqualified.
6	System	Blocks promotion/awards.
7	System	Writes the audit log.
8	System	Sends notification to Team Members.

9	System	Recalculates ranking.
10	Coordinator / System	If disqualification is revoked, recalculates and returns the entry to Pending.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Result editable?	No: reject change. Yes: continue with Disqualified, audit, notification, and recalculation.
Source state	Pending, Qualified, or Eliminated can transition to Disqualified.
Revoke disqualification	Disqualified -> Pending followed by recalculation.

1.9 Result Publishing and Reporting

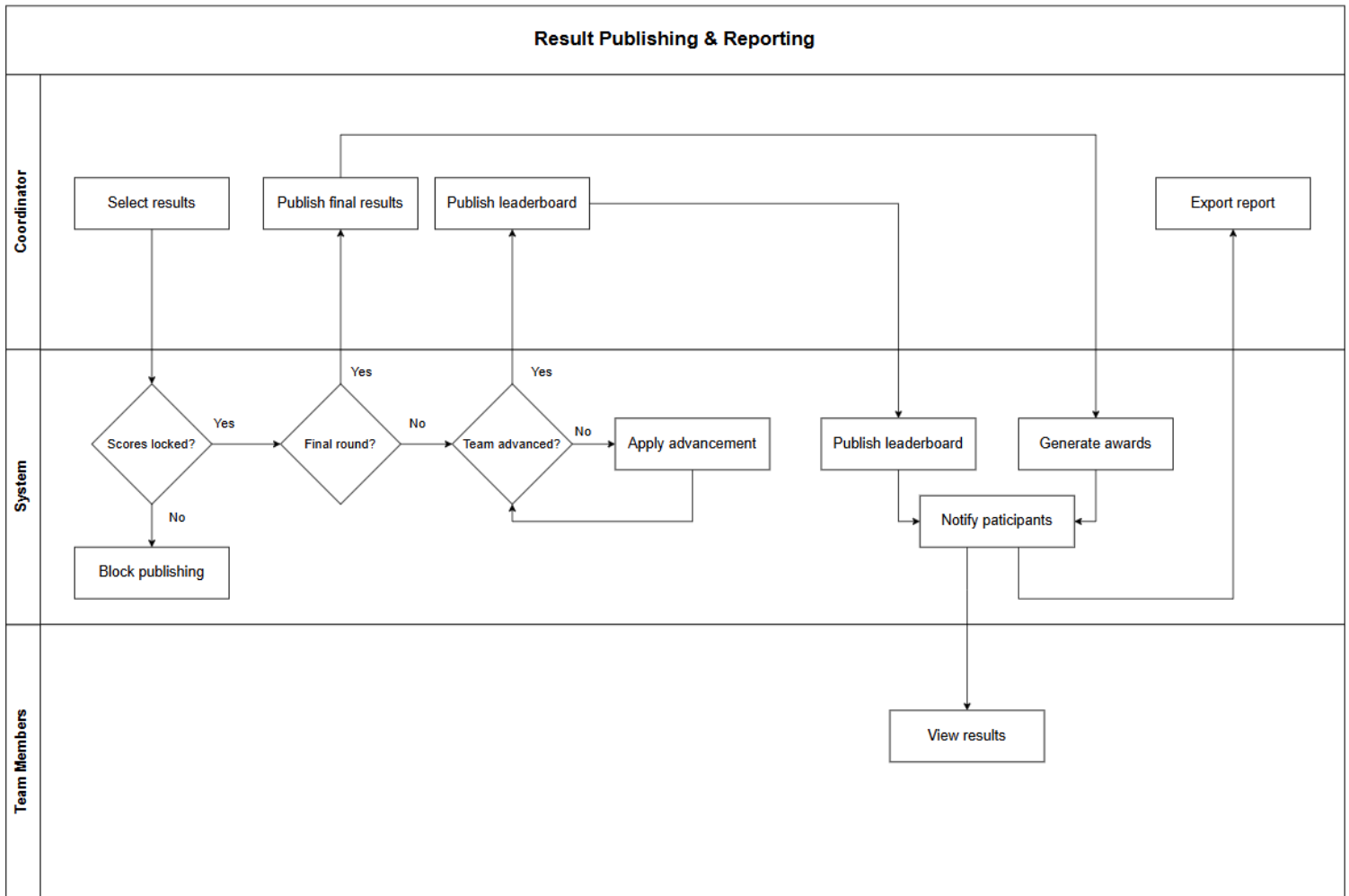


Figure 1.9 Result Publishing and Reporting

Process boundary

Field	Description
Business-flow code	BF-09
Purpose	Publish locked non-final or final results, ensure advancement before a non-final leaderboard, create final awards, notify participants, and support result viewing/export.
Inputs	Selected results, scores-locked status, final/non-final round type, advancement state, ranking/leaderboard, and final award trigger.

Start event	The coordinator selects results for publication; the system is in Unpublished.
End event	The result is RoundPublished or FinalPublished/AwardsGenerated; participants are notified and results are viewable/exportable.
Outputs	RoundPublished leaderboard or FinalPublished result, publishedAt, AwardsGenerated for final results, participant notifications, viewed results, and exported report.

Lane responsibilities

Lane / actor	Responsibility
Coordinator	Selects results, publishes final results or leaderboard, and exports the report.
System	Checks publishing guards, applies advancement, publishes, creates awards, and notifies participants.
Team Member	Receive notifications and view results.

Main flow

Step	Responsible lane	Activity / expected result
1	Coordinator	Selects results.
2	System	Checks Scores locked?.
3	System	If locked, checks Final round?.
4	System	For a non-final round, checks Team advanced?.
5	System	If advancement is missing, applies advancement and checks again.
6	Coordinator / System	Publishes the leaderboard after advancement; the state becomes RoundPublished.
7	Coordinator	For the final round, publishes final results.
8	System	Sets publishedAt and changes Unpublished to FinalPublished.
9	System	Automatically creates awards and changes FinalPublished to AwardsGenerated.
10	System	Notifies participants.

11	Team Member	View results.
12	Coordinator	Exports the report.

Decision points and exception flow

Decision / exception	Branch, handling and outcome
Scores locked?	No: block publishing. Yes: continue.
Final round?	No: use the leaderboard/advancement path. Yes: publish final results.
Team advanced?	No: apply advancement before publishing the non-final leaderboard. Yes: publish.
Award trigger	Awards are generated automatically only after FinalPublished.

2. Business Rules

1. Authentication

ID	Business Rule	Acceptance Criteria
BR1-01	Only email/password and Google authentication are supported.	Users can authenticate using either email/password or Google.
BR1-02	Only active accounts can log in.	Inactive accounts cannot access the system.
BR1-03	Passwords must meet the password policy.	Passwords that violate the policy are rejected.
BR1-04	Email addresses must be verified before registration.	Registration is completed only after successful OTP verification.

2. Account Management

ID	Business Rule	Acceptance Criteria
BR2-01	Only Coordinators can approve accounts.	Only Coordinators can approve or reject pending accounts.
BR2-02	Password reset requires OTP verification.	Password reset is allowed only after OTP verification.
BR2-03	OTP must be valid.	Expired or invalid OTPs are rejected.
BR2-04	Current passwords must be verified before changes.	Password updates fail if the current password is incorrect.
BR2-05	New passwords must differ from current passwords.	Reusing the current password is not allowed.

3. Event Management

ID	Business Rule	Acceptance Criteria
BR3-01	Only Coordinators can manage events.	Only Coordinators can create, update, or delete events.
BR3-02	Only one event is allowed per season each year.	Duplicate events for the same season and year are rejected.
BR3-03	An event must contain at least one track and one round.	Events cannot be published without a track and a round.
BR3-04	Event schedules must be valid.	Invalid event dates or round deadlines are rejected.
BR3-05	Criteria weights must total 100%.	Criteria cannot be saved unless the total weight equals 100%.

4. Team Management

ID	Business Rule	Acceptance Criteria
BR4-01	Teams must contain 3–5 members.	Teams outside this range cannot participate.
BR4-02	Only students can join teams.	Non-student users cannot join a team.
BR4-03	Team leadership transfer requires recipient acceptance.	Leadership changes only after the recipient accepts the request.
BR4-04	Team leaders cannot leave before transferring leadership.	Team leaders cannot leave unless leadership has been transferred.

5. Submission Management

ID	Business Rule	Acceptance Criteria
BR5-01	Only registered teams can submit projects.	Unregistered teams cannot submit projects.
BR5-02	Submissions must be made before the deadline.	Late submissions are rejected.
BR5-03	Required submission information must be provided.	Incomplete submissions cannot be submitted.
BR5-04	Submissions can be updated before the deadline only.	Submission editing is disabled after the deadline.

6. Judge Management

ID	Business Rule	Acceptance Criteria
BR6-01	Only Coordinators can assign judges.	Only Coordinators can assign judges to rounds.
BR6-02	Judges evaluate assigned submissions only.	Judges cannot access unassigned submissions.

BR6-03	Guest judges cannot self-register.	Guest judge accounts are created by Coordinators only.
BR6-04	A lecturer can hold both Mentor and Judge roles if assigned to different tracks.	Dual-role assignments are allowed only across different tracks.

7. Evaluation

ID	Business Rule	Acceptance Criteria
BR7-01	Evaluations follow predefined criteria.	Judges evaluate submissions using the assigned scoring rubric.
BR7-02	Each judge submits an independent evaluation.	Each judge's evaluation is recorded independently.
BR7-03	Scores are locked after round closure.	Closed-round scores cannot be modified.
BR7-04	Judges and mentors can provide feedback.	Teams can view submitted feedback.
BR7-05	Feedback history is preserved.	Previous feedback remains accessible.

8. Ranking & Awards

ID	Business Rule	Acceptance Criteria
BR8-01	Rankings are based on evaluation results.	Rankings are generated from final evaluation scores.
BR8-02	Leaderboards are available after publication only.	Users can view the leaderboard only after publication.
BR8-03	Team promotion follows competition rules.	Only qualified teams advance to the next round.
BR8-04	Awards are granted based on final rankings.	Awards are assigned according to the published rankings.

BR8-05	Only Coordinators can disqualify teams.	Only Coordinators can disqualify teams or submissions.
BR8-06	Disqualified teams are not eligible for awards.	Disqualified teams are excluded from rankings and awards.

9. Report & Audit

ID	Business Rule	Acceptance Criteria
BR9-01	Only Coordinators can generate reports.	Reports are available only to Coordinators.
BR9-02	Reports include ranking and scoring data.	Generated reports contain ranking and evaluation results.
BR9-03	Important system activities must be logged.	Audit logs record significant user actions.

10. Communication

ID	Business Rule	Acceptance Criteria
BR10-01	Only Coordinators can publish announcements.	Published announcements are visible to participants.
BR10-02	Notify users of important activities.	Notifications are sent for relevant system events.

3. State Chart

This section specifies the lifecycle of nine important SEAL business objects. Each chart definition identifies the initial/final state, state meanings, transition trigger or event, guard condition, and action produced by the transition. A replaceable landscape placeholder follows every specification.

3.1 Account Registration and Approval State

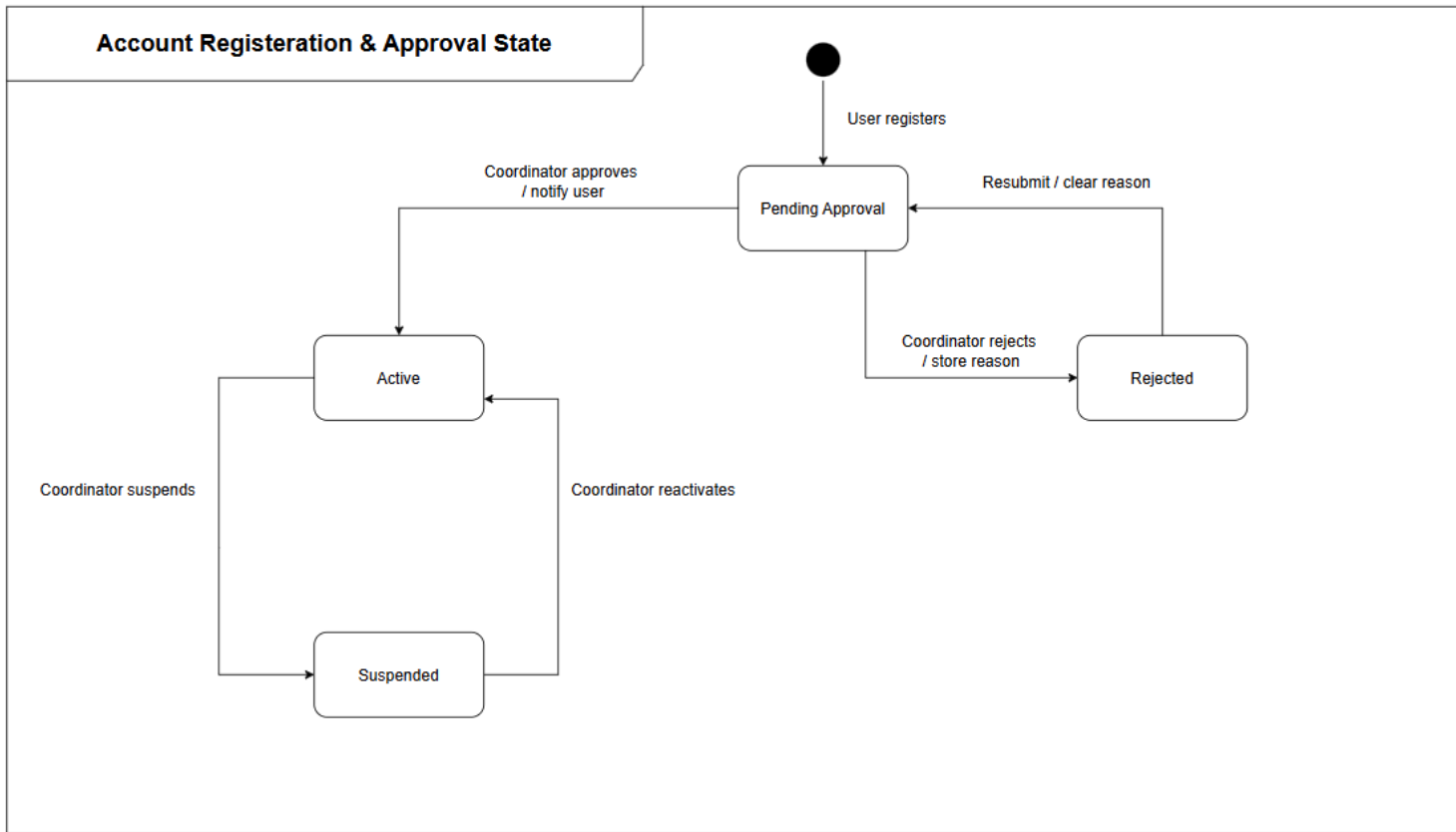


Figure 3.1 Account Registration and Approval State Chart

Lifecycle boundary

Field	Description
Business object	Account
Purpose	Describe the account registration and approval lifecycle, including rejection, resubmission, suspension, and reactivation.
Actors	User, Coordinator, and System
Initial state	Initial pseudostate -> Pending Approval
Final / terminal state	No explicit final pseudostate is drawn; Active, Rejected, and Suspended are persistent/recoverable states.

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before registration is recorded.
Pending Approval	Business state	Registration is awaiting coordinator review.
Active	Business state	The approved account can use authorized system functions.
Rejected	Business state	The application was rejected and stores a reason until resubmission.
Suspended	Business state	An Active account has been temporarily disabled by the coordinator.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	User registers	Pending Approval
T2	Pending Approval	Coordinator approves	Active
T3	Pending Approval	Coordinator rejects	Rejected
T4	Rejected	Resubmit	Pending Approval
T5	Active	Coordinator suspends	Suspended
T6	Suspended	Coordinator reactivates	Active

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	None specified	Notify the user.
T3	None specified	Store the rejection reason.

T4	None specified	Clear the rejection reason.
T5	None specified	No action specified in the diagram.
T6	None specified	No action specified in the diagram.

3.2 Event Lifecycle State

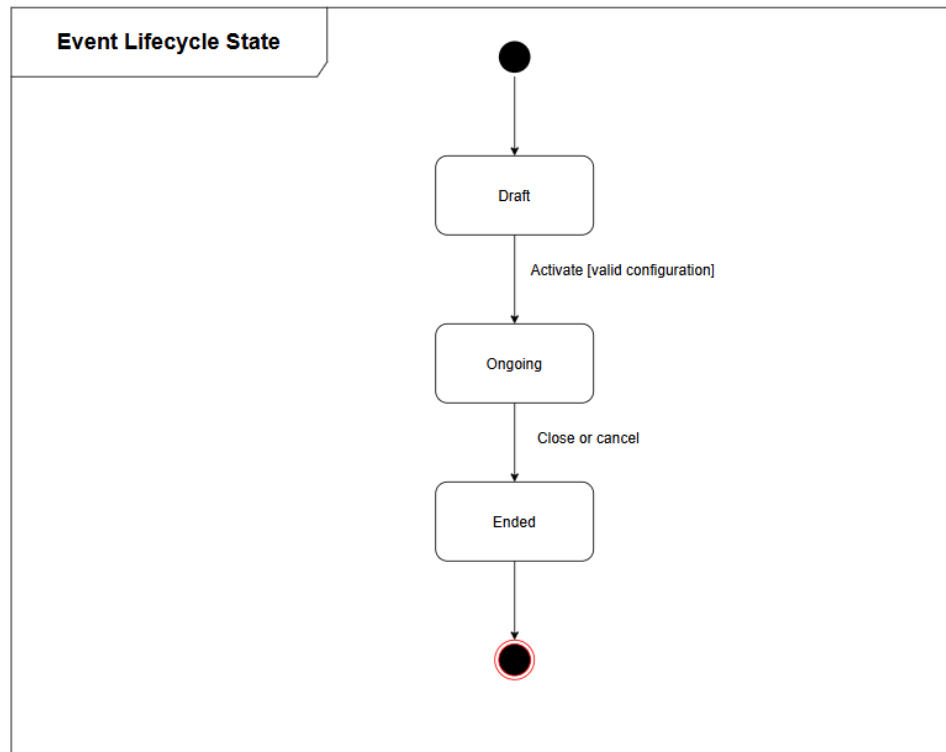


Figure 3.2 Event Lifecycle State Chart

Lifecycle boundary

Field	Description
Business object	Event
Purpose	Describe the event lifecycle from creation as Draft through activation, closure/cancellation, and completion.
Actors	Coordinator and System
Initial state	Initial pseudostate -> Draft
Final / terminal state	Ended -> final pseudostate

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before the event record is created.
Draft	Business state	The event exists but has not been activated; configuration can be completed and validated.
Ongoing	Business state	The event is active after configuration validation.
Ended	Terminal business state	The active event has been closed or cancelled.
Final	Final pseudostate	Lifecycle processing is complete after Ended.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Unlabeled initial transition	Draft
T2	Draft	Activate	Ongoing
T3	Ongoing	Close or cancel	Ended
T4	Ended	Unlabeled completion transition	Final

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	Valid configuration	No action specified in the diagram.
T3	None specified	No action specified in the diagram.
T4	None specified	No action specified in the diagram.

3.3 Team State

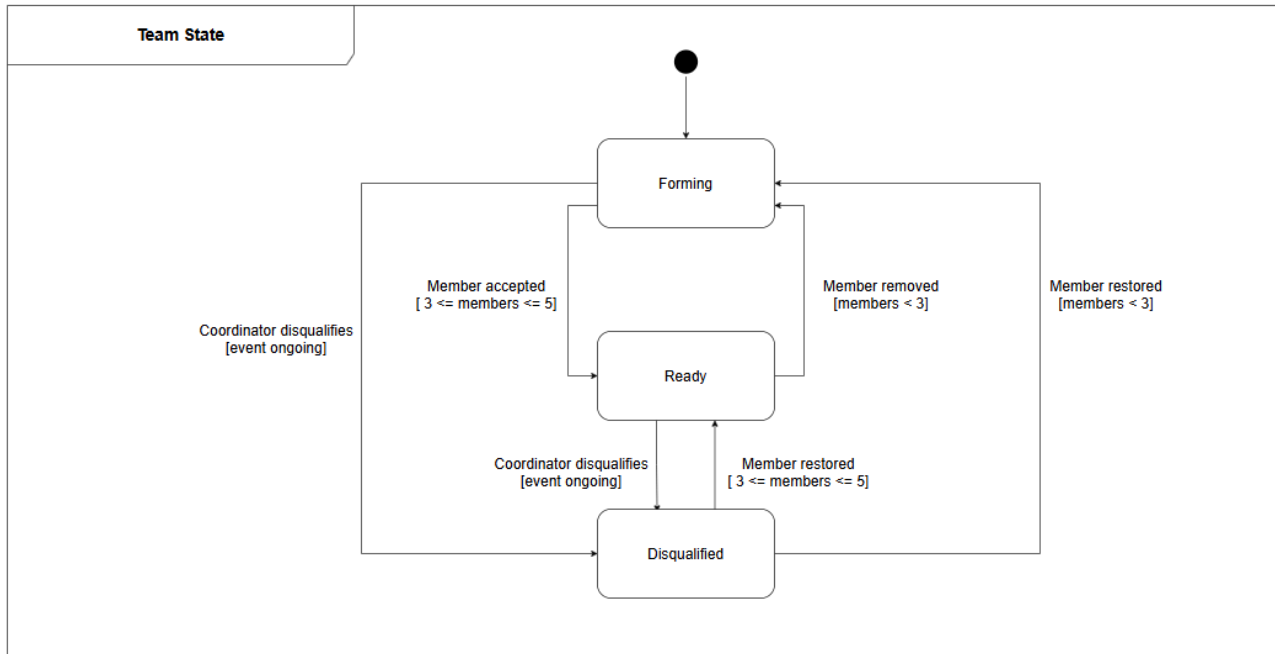


Figure 3.3 Team State Chart

Lifecycle boundary

Field	Description
Business object	Team
Purpose	Describe how team membership and coordinator disqualification move a team among Forming, Ready, and Disqualified.
Actors	Team Leader, Team Member, Coordinator, and System
Initial state	Initial pseudostate -> Forming
Final / terminal state	No explicit final pseudostate is drawn; Forming, Ready, and Disqualified remain recoverable business states.

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point when a team is created.
Forming	Business state	The team exists but has fewer than the required three members or is still being assembled.

Ready	Business state	The team has between three and five members.
Disqualified	Business state	The coordinator has disqualified the team while the event is ongoing.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Unlabeled initial transition	Forming
T2	Forming	Member accepted	Ready
T3	Ready	Member removed	Forming
T4	Forming	Coordinator disqualifies	Disqualified
T5	Ready	Coordinator disqualifies	Disqualified
T6	Disqualified	Member restored	Forming
T7	Disqualified	Member restored	Ready

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	$3 \leq \text{members} \leq 5$	No action specified in the diagram.
T3	$\text{members} < 3$	No action specified in the diagram.
T4	Event ongoing	No action specified in the diagram.
T5	Event ongoing	No action specified in the diagram.
T6	$\text{members} < 3$	No action specified in the diagram.
T7	$3 \leq \text{members} \leq 5$	No action specified in the diagram.

3.4 Staff Assignment State

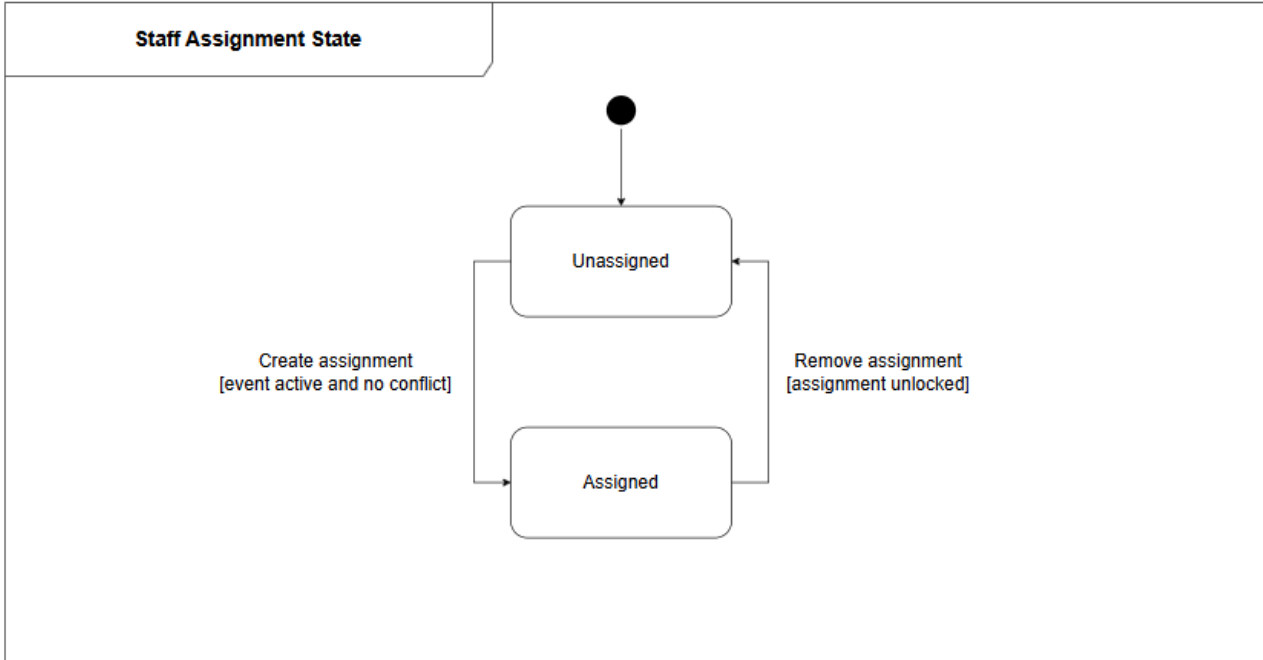


Figure 3.4 Staff Assignment State Chart

Lifecycle boundary

Field	Description
Business object	Mentor/Judge Assignment
Purpose	Describe creation and removal of a Mentor/Judge assignment with active-event, conflict, and lock guards.
Actors	Coordinator, Mentor, Judge, and SEAL System
Initial state	Initial pseudostate -> Unassigned
Final / terminal state	No explicit final pseudostate is drawn; the object alternates between Unassigned and Assigned.

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before an assignment exists.
Unassigned	Business state	No active Mentor/Judge assignment exists for the selected scope.

Assigned	Business state	A conflict-free assignment has been created for an active event.
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Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Unlabeled initial transition	Unassigned
T2	Unassigned	Create assignment	Assigned
T3	Assigned	Remove assignment	Unassigned

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	Event active and no conflict	No action specified in the diagram.
T3	Assignment unlocked	No action specified in the diagram.

3.5 Submission State

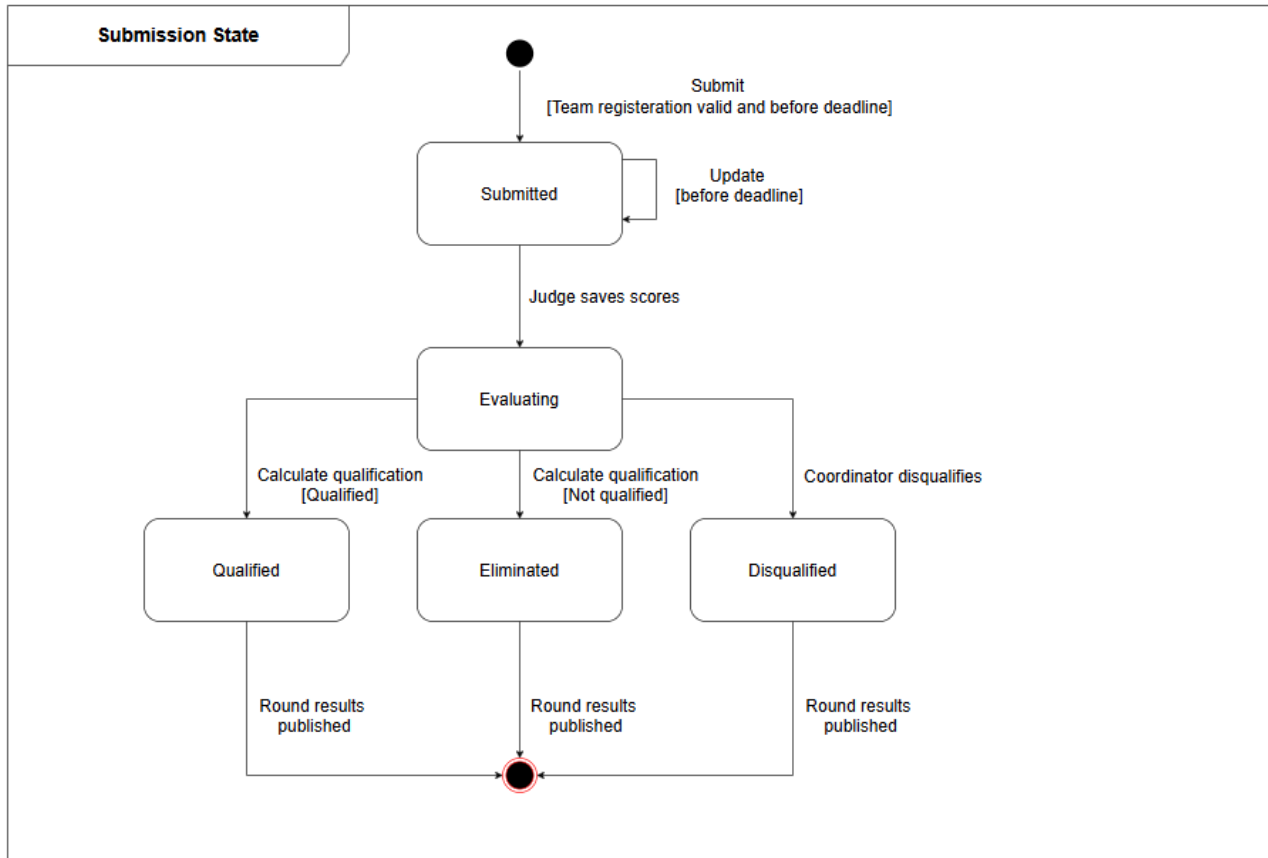


Figure 3.5 Submission State Chart

Lifecycle boundary

Field	Description
Business object	Round Submission
Purpose	Describe initial submission, pre-deadline updates, transition into evaluation, qualification outcomes, disqualification, and published completion.
Actors	Team Leader, Judge, Coordinator, and System
Initial state	Initial pseudostate -> Submitted
Final / terminal state	Qualified, Eliminated, or Disqualified -> final pseudostate after round results are published.

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before a team submits round links.
Submitted	Business state	A valid on-time submission has been stored and may still be updated before the deadline.
Evaluating	Business state	A Judge has saved scores for the submission.
Qualified	Outcome state	Qualification calculation marked the submission as qualified.
Eliminated	Outcome state	Qualification calculation marked the submission as not qualified.
Disqualified	Outcome state	The coordinator disqualified the submission/team.
Final	Final pseudostate	Round results have been published for the outcome.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Submit	Submitted
T2	Submitted	Update	Submitted
T3	Submitted	Judge saves scores	Evaluating
T4	Evaluating	Calculate qualification	Qualified
T5	Evaluating	Calculate qualification	Eliminated
T6	Evaluating	Coordinator disqualifies	Disqualified
T7	Qualified	Round results published	Final
T8	Eliminated	Round results published	Final
T9	Disqualified	Round results published	Final

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	Team registration valid and before deadline	No action specified in the diagram.
T2	Before deadline	No action specified in the diagram.
T3	None specified	No action specified in the diagram.
T4	Qualified	No action specified in the diagram.
T5	Not qualified	No action specified in the diagram.
T6	None specified	No action specified in the diagram.
T7	None specified	No action specified in the diagram.
T8	None specified	No action specified in the diagram.
T9	None specified	No action specified in the diagram.

3.6 Judge Evaluation State

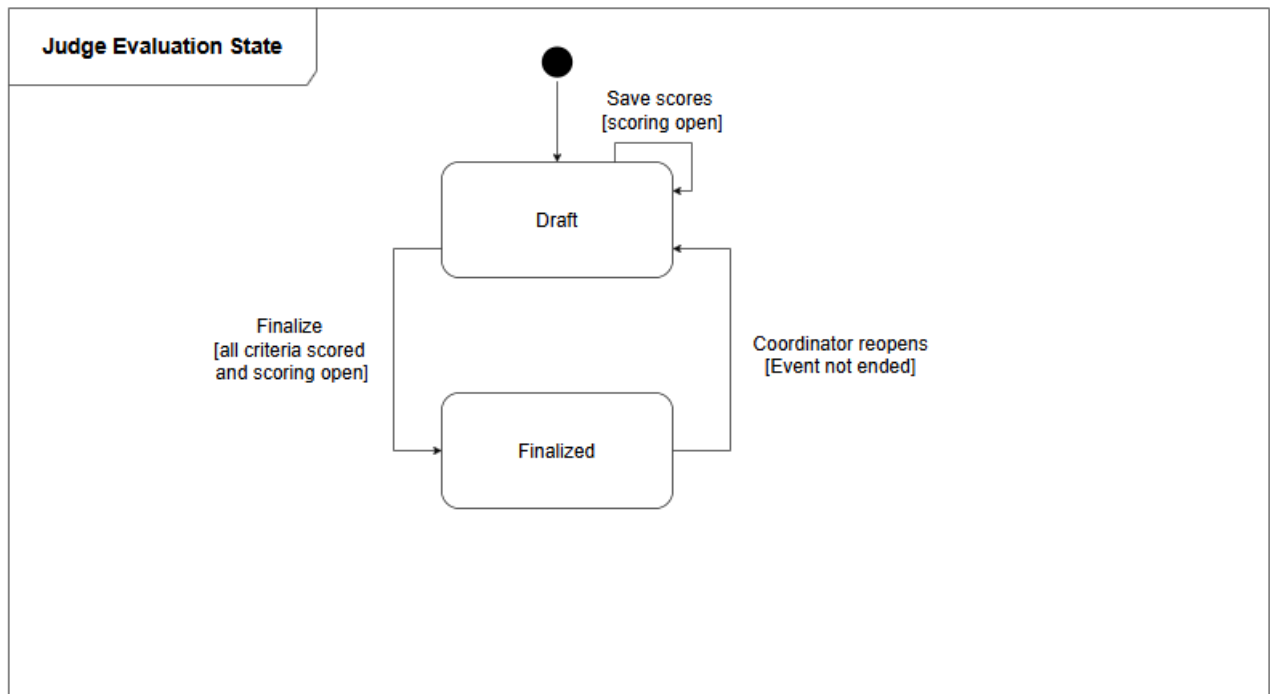


Figure 3.6 Judge Evaluation State Chart

Lifecycle boundary

Field	Description
Business object	Judge Evaluation
Purpose	Describe Draft scoring, repeated score saves, finalization, and coordinator reopening of a Judge evaluation.
Actors	Judge, Coordinator, and System
Initial state	Initial pseudostate -> Draft when scores are first saved
Final / terminal state	No explicit final pseudostate is drawn; Finalized can return to Draft through reopening.

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before the Judge has saved any score.
Draft	Business state	Scores/feedback have been saved but the evaluation is still editable.
Finalized	Business state	All criteria are scored and the Judge has finalized while scoring is open.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Save scores	Draft
T2	Draft	Save scores	Draft
T3	Draft	Finalize	Finalized
T4	Finalized	Coordinator reopens	Draft

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	Scoring open	No action specified in the diagram.

T2	Scoring open	No action specified in the diagram.
T3	All criteria scored and scoring open	No action specified in the diagram.
T4	Event not ended	No action specified in the diagram.

3.7 Round Result Processing State

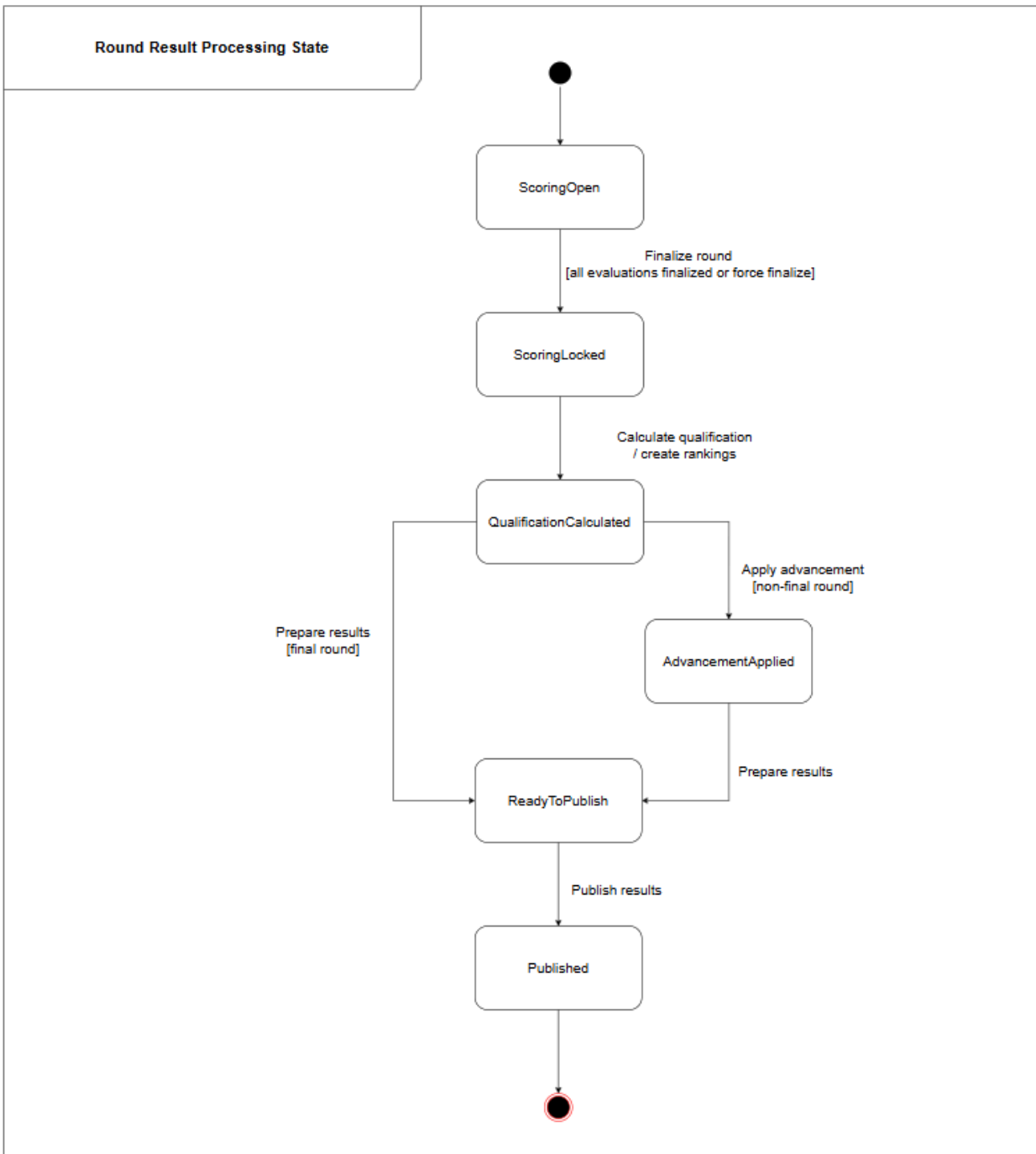


Figure 3.7 Round Result Processing State Chart

Lifecycle boundary

Field	Description
Business object	Round Result Processing
Purpose	Describe score locking, ranking/qualification calculation, advancement, preparation, publication, and completion of a round result.
Actors	Coordinator and System
Initial state	Initial pseudostate -> ScoringOpen
Final / terminal state	Published -> final pseudostate

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point for round-result processing.
ScoringOpen	Business state	Judge evaluations may still be saved/finalized.
ScoringLocked	Business state	Round scoring has been finalized or force-finalized and is locked.
QualificationCalculated	Business state	Ranking and qualification have been calculated.
AdvancementApplied	Business state	Non-final round advancement has been applied.
ReadyToPublish	Business state	All required preparation for publication is complete.
Published	Terminal business state	The round result has been published.
Final	Final pseudostate	Round-result processing is complete.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Unlabeled initial transition	ScoringOpen
T2	ScoringOpen	Finalize round	ScoringLocked
T3	ScoringLocked	Calculate qualification	QualificationCalculated
T4	QualificationCalculated	Apply advancement	AdvancementApplied
T5	QualificationCalculated	Prepare results	ReadyToPublish
T6	AdvancementApplied	Prepare results	ReadyToPublish
T7	ReadyToPublish	Publish results	Published
T8	Published	Unlabeled completion transition	Final

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	All evaluations finalized or force finalize	No action specified in the diagram.
T3	None specified	Create rankings.
T4	Non-final round	No action specified in the diagram.
T5	Final round	No action specified in the diagram.
T6	None specified	No action specified in the diagram.
T7	None specified	No action specified in the diagram.
T8	None specified	No action specified in the diagram.

3.8 Ranking Qualification State

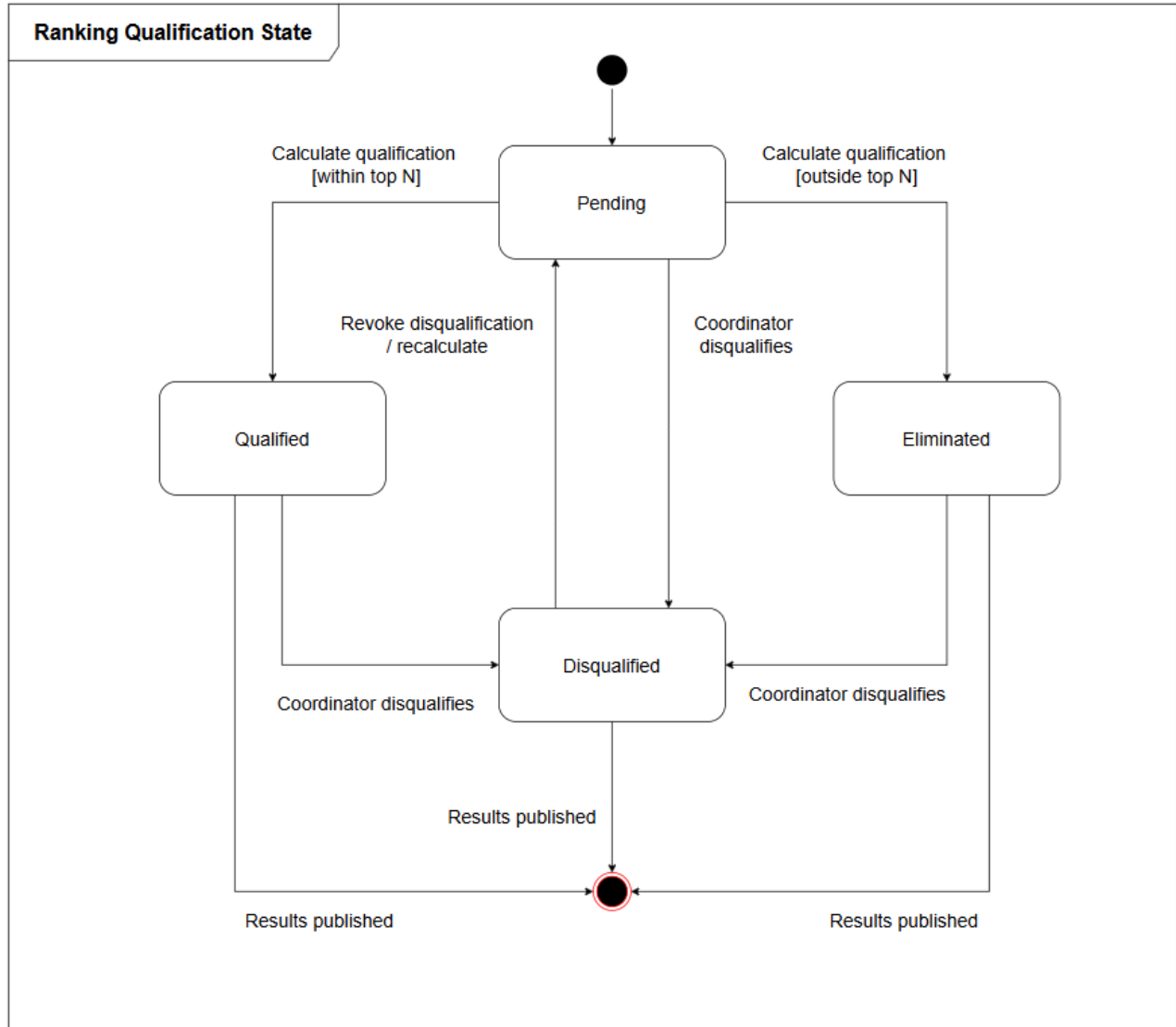


Figure 3.8 Ranking Qualification State Chart

Lifecycle boundary

Field	Description
Business object	Ranking Qualification
Purpose	Describe top-N qualification, elimination, disqualification, revocation/recalculation, and completion after results publication.
Actors	Coordinator and System
Initial state	Initial pseudostate -> Pending

Final / terminal state	Qualified, Eliminated, or Disqualified -> final pseudostate when results are published.
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State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before qualification is calculated.
Pending	Business state	The ranking entry awaits qualification calculation or recalculation.
Qualified	Outcome state	The entry is within top N.
Eliminated	Outcome state	The entry is outside top N.
Disqualified	Outcome state	The coordinator disqualified the entry.
Final	Final pseudostate	The outcome is complete after result publication.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Unlabeled initial transition	Pending
T2	Pending	Calculate qualification	Qualified
T3	Pending	Calculate qualification	Eliminated
T4	Pending	Coordinator disqualifies	Disqualified
T5	Qualified	Coordinator disqualifies	Disqualified
T6	Eliminated	Coordinator disqualifies	Disqualified
T7	Disqualified	Revoke disqualification	Pending
T8	Qualified	Results published	Final
T9	Eliminated	Results published	Final
T10	Disqualified	Results published	Final

Guard conditions and actions

Transiti on	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	Within top N	No action specified in the diagram.
T3	Outside top N	No action specified in the diagram.
T4	None specified	No action specified in the diagram.
T5	None specified	No action specified in the diagram.
T6	None specified	No action specified in the diagram.
T7	None specified	Recalculate.
T8	None specified	No action specified in the diagram.
T9	None specified	No action specified in the diagram.
T10	None specified	No action specified in the diagram.

3.9 Result Publication State

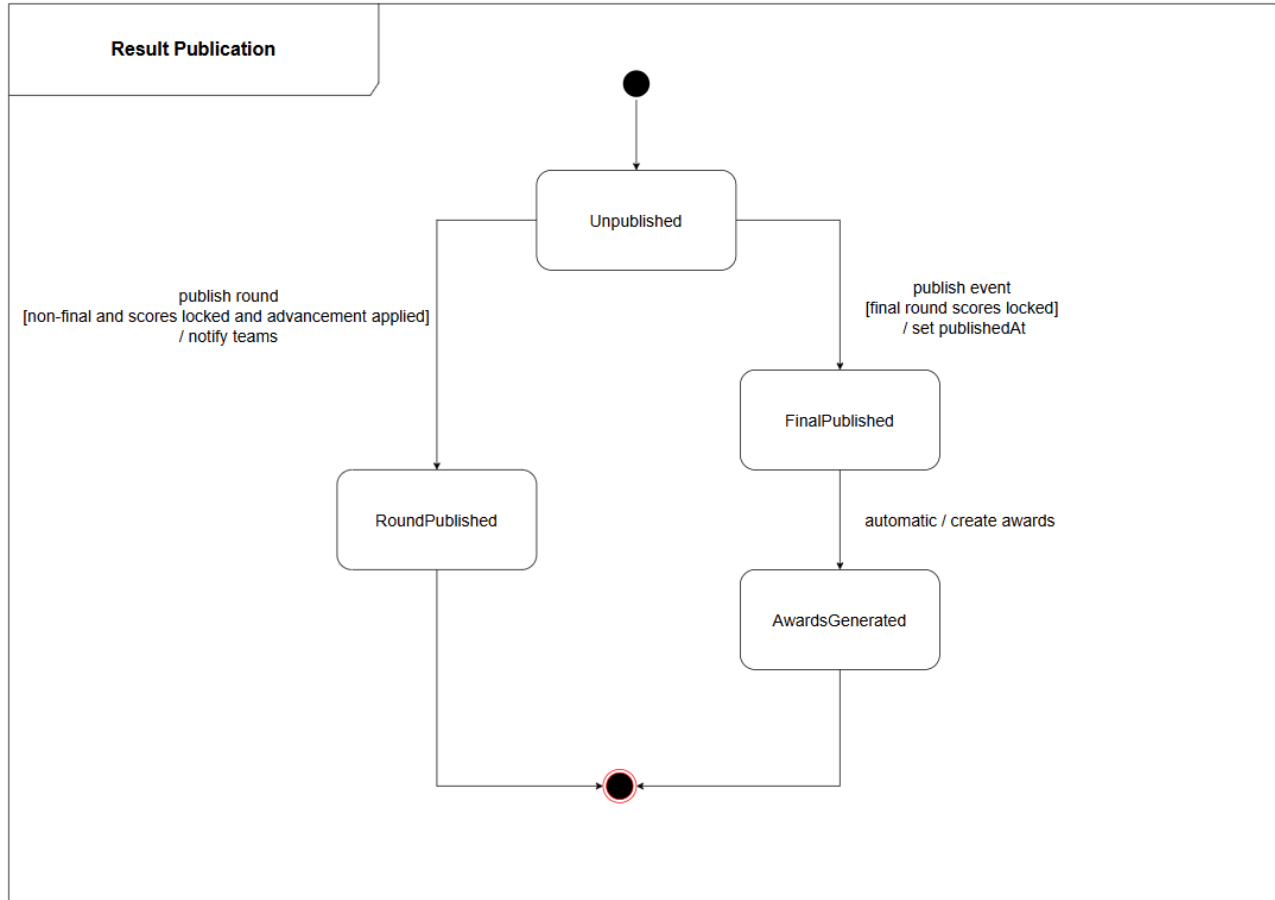


Figure 3.9 Result Publication State Chart

Lifecycle boundary

Field	Description
Business object	Result Publication
Purpose	Describe guarded publication of non-final and final results, notification, publishedAt, automatic award creation, and completion.
Actors	Coordinator, Team Members, and System
Initial state	Initial pseudostate -> Unpublished
Final / terminal state	RoundPublished or AwardsGenerated -> final pseudostate

State catalogue

State	Type	Meaning
Initial	Initial pseudostate	Entry point before result publication is prepared.
Unpublished	Business state	The result exists but publication guards have not yet produced a published outcome.
RoundPublished	Terminal business state	A non-final result has been published after scores are locked and advancement is applied.
FinalPublished	Business state	The final event result has been published and publishedAt is set.
AwardsGenerated	Terminal business state	Awards were created automatically after final publication.
Final	Final pseudostate	Publication processing is complete.

Transition matrix

ID	From state	Trigger / event	To state
T1	Initial	Unlabeled initial transition	Unpublished
T2	Unpublished	Publish round	RoundPublished
T3	Unpublished	Publish event	FinalPublished
T4	FinalPublished	Automatic	AwardsGenerated
T5	RoundPublished	Unlabeled completion transition	Final
T6	AwardsGenerated	Unlabeled completion transition	Final

Guard conditions and actions

Transition	Guard condition	Action / side effect
T1	None specified	No action specified in the diagram.
T2	Non-final and scores locked and advancement applied	Notify teams.

T3	Final round scores locked	Set publishedAt.
T4	None specified	Create awards.
T5	None specified	No action specified in the diagram.
T6	None specified	No action specified in the diagram.

4. Context Diagram

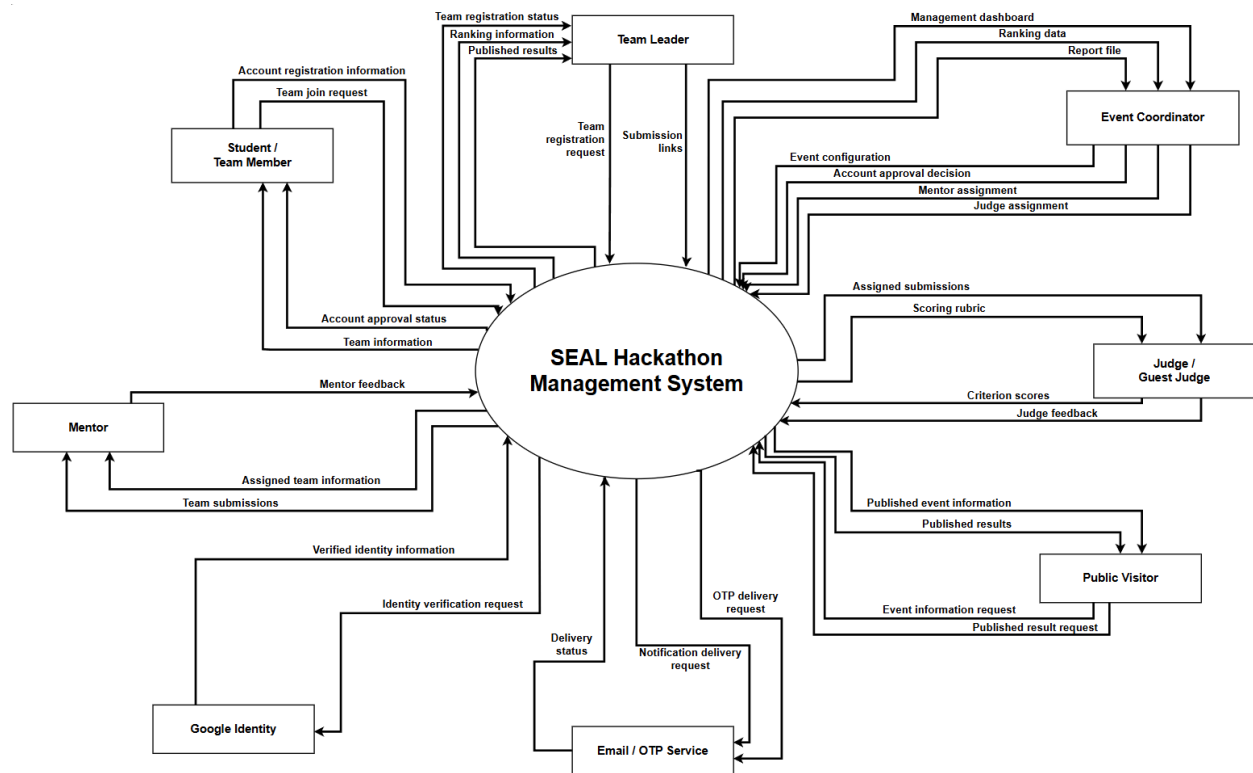


Figure 4.1 SEAL system context diagram

5. Conceptual Model

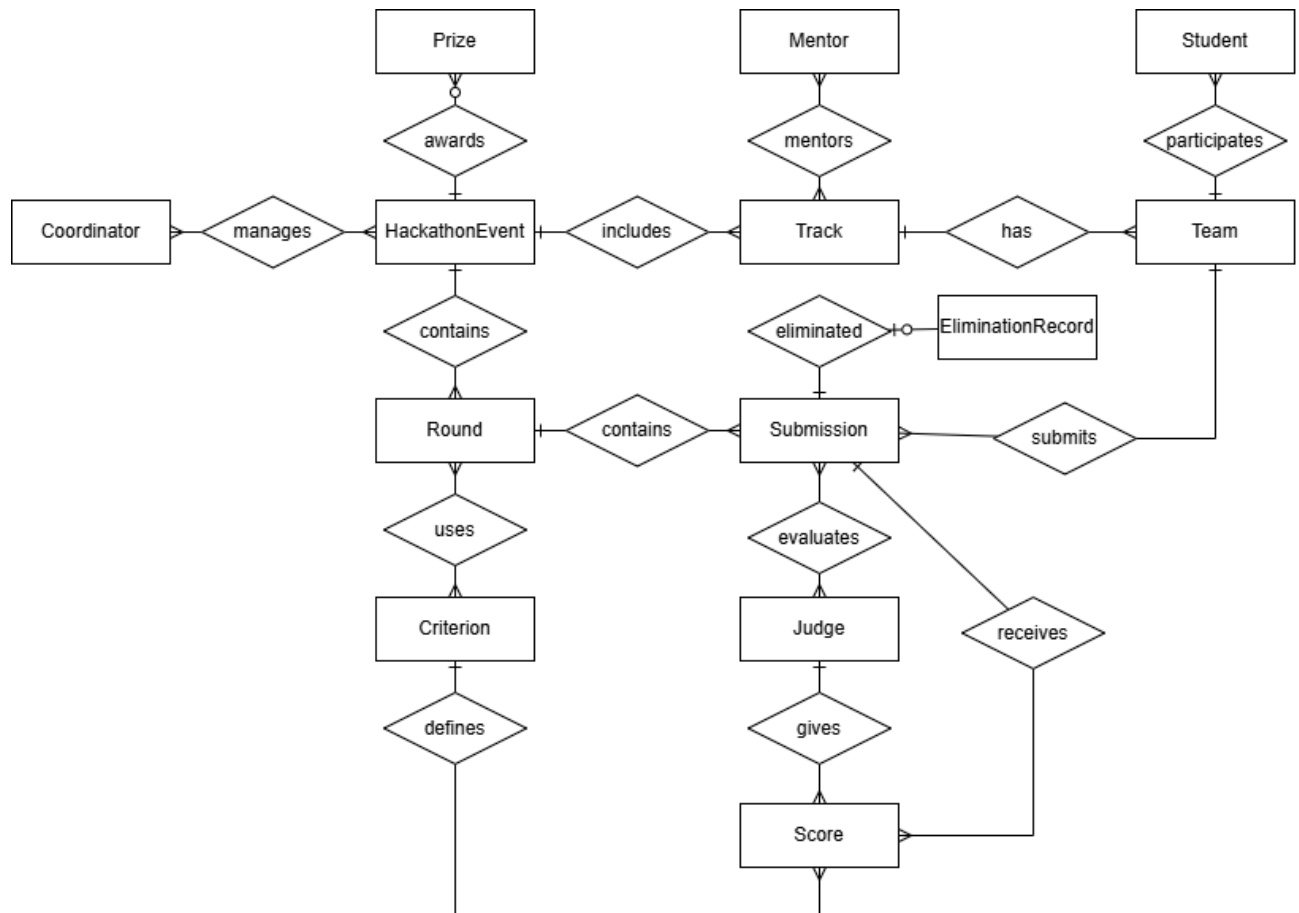


Figure 5.1 SEAL conceptual model

6. Entity Relationship Diagram (ERD)

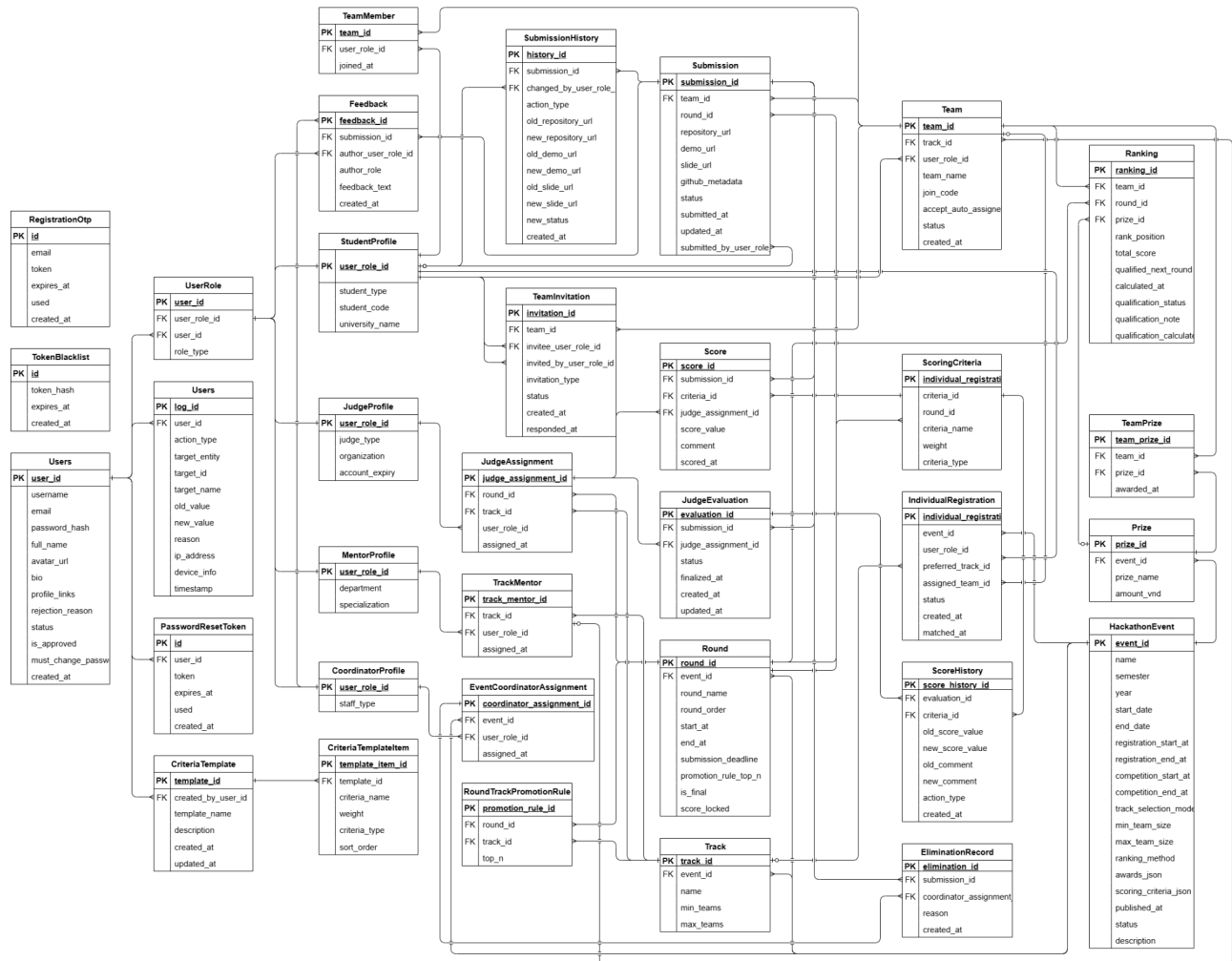


Figure 6.1 SEAL entity relationship diagram